Practice workbook · Finding n in a geometric series with logarithms

Setting up an equation or inequality for a term or sum of a geometric series, isolating r^n and taking logarithms to find n.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-14

1. Question 8 · part d · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.

8 The sum of the first and second terms of a geometric series G is 400

The sum of the second and third terms of G is 100

(a) Show that the common ratio of G is $\frac{1}{4}$

(4)

(b) Show that the first term of G is 320

(2)

(c) Find the sum to infinity of G

(2)

The sum to n terms of G is S_n

(d) Find, using logarithms, the least value of n such that

$$S_n > 426.6$$

(4)

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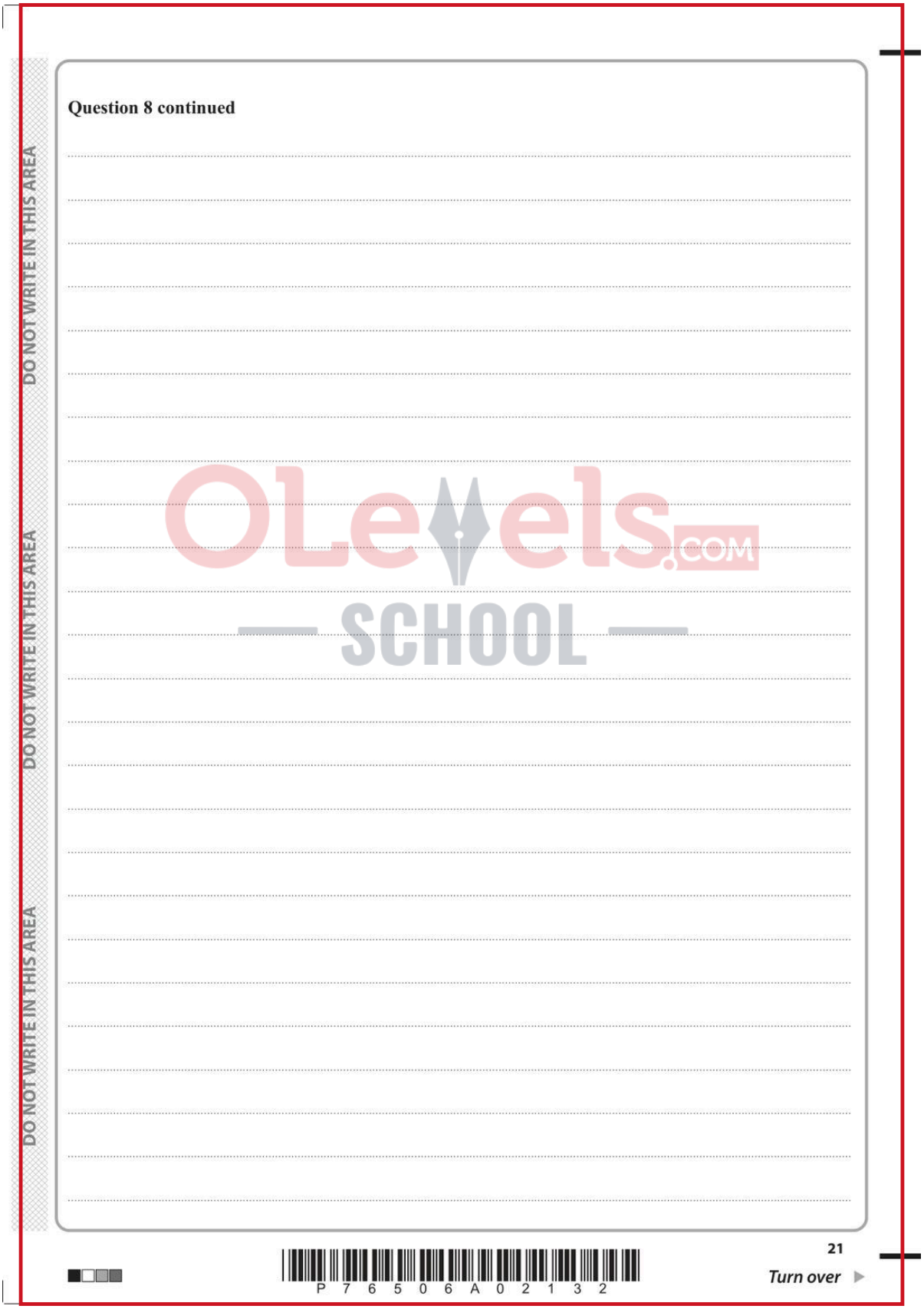
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1. Question 8 · part d · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.



1. Question 8 · part d · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.

Question 8 continued

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1. Question 8 · part d · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part d only.

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Question 8 continued

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(Total for Question 8 is 12 marks)

1. Question 8 · part d · 4 marks · Working and answer

Working and answer

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2. Question 7 · part b · 4 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

7 The nth term of a geometric series G is u_n

The first term of G is a and the common ratio of G is r, where $r > 0$

Given that $u_3 = 4$ and that $u_7 = 16$

(a) (i) show that $r = \sqrt{2}$

(ii) find the value of a.

(3)

(b) Find the least value of n for which $u_n > 500$

(4)

The sum of the first n terms of G is S_n

(c) Find S_{20}

Give your answer in the form $p\left(1+\sqrt{2}\right)$ where p is an integer.

(4)

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2. Question 7 · part b · 4 marks · Working and answer

Working and answer

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3. Question 9 · part b · 3 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

9 A geometric series G has common ratio r where $r > 0$

The third term of G is $\frac{27}{2}$ and the sum of the first three terms of G is $\frac{57}{2}$

Given that the sum to n terms of G is S_n

(a) show that $S_n = \sum_{j=1}^{n} 4\left(\frac{3}{2}\right)^j$

(8)

Given that $S_k > 50\,000$

(b) show that the least value of k is given by

$$k > \frac{\lg\left(\frac{12\,503}{3}\right)}{\lg\left(\frac{3}{2}\right)}$$

(3)

(c) Hence find the least value of k

(1)

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3. Question 9 · part b · 3 marks · Working and answer

Working and answer

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4. Question 9 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

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4. Question 9 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 9 continued

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4. Question 9 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 9 continued

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4. Question 9 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.



4. Question 9 · part c · 1 mark · Working and answer

Working and answer

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5. Question 7 · part d · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part d only.

7 A geometric series has first term $(x - 3)$, second term $(x + 1)$ and third term $(4x - 2)$.

(a) Find the two possible values of x.

(5)

Given that $x < 1$,

(b) show that the series is convergent.

(2)

The sum to infinity of the series is S.

(c) Find the value of S.

(2)

The sum of the first n terms of the series is S_n

Given that $\frac{S}{S_n} = \frac{256}{255}$

(d) find the value of n.

(3)

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5. Question 7 · part d · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part d only.

Question 7 continued

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5. Question 7 · part d · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part d only.

Question 7 continued

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5. Question 7 · part d · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part d only.



Question 7 continued

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(Total for Question 7 is 12 marks)

5. Question 7 · part d · 3 marks · Working and answer

Working and answer

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6. Question 8 · part c · 6 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

8 The nth term of a geometric series G is U_n and the sum of the first n terms of G is S_n

Given that $U_n = \frac{25}{4}\left(\frac{3}{5}\right)^n$

(a) find the exact value of U_5

(1)

(b) Show that $S_n = \sum_{r=1}^{n} \frac{A}{B}\left(\frac{3}{5}\right)^{r-1}$ where A and B are integers to be found.

(3)

The sum to infinity of G is S

(c) Find the least value of n such that $S - S_n < 0.045$

(6)

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6. Question 8 · part c · 6 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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6. Question 8 · part c · 6 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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6. Question 8 · part c · 6 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 10 marks)

6. Question 8 · part c · 6 marks · Working and answer

Working and answer

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7. Question 8 · part d · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

8 The sum of the first 2 terms of a geometric series G is 360
The sum of the 2nd and 3rd terms of G is 288

The nth term of G is U_n

(a) Show that $U_n = A\left(\frac{4}{5}\right)^{n-1}$ where A is an integer to be found.

(7)

(b) Explain why G is convergent.

(1)

(c) Hence find the sum to infinity of G

(2)

(d) Find the least number of terms for which the sum is greater than 978

(4)

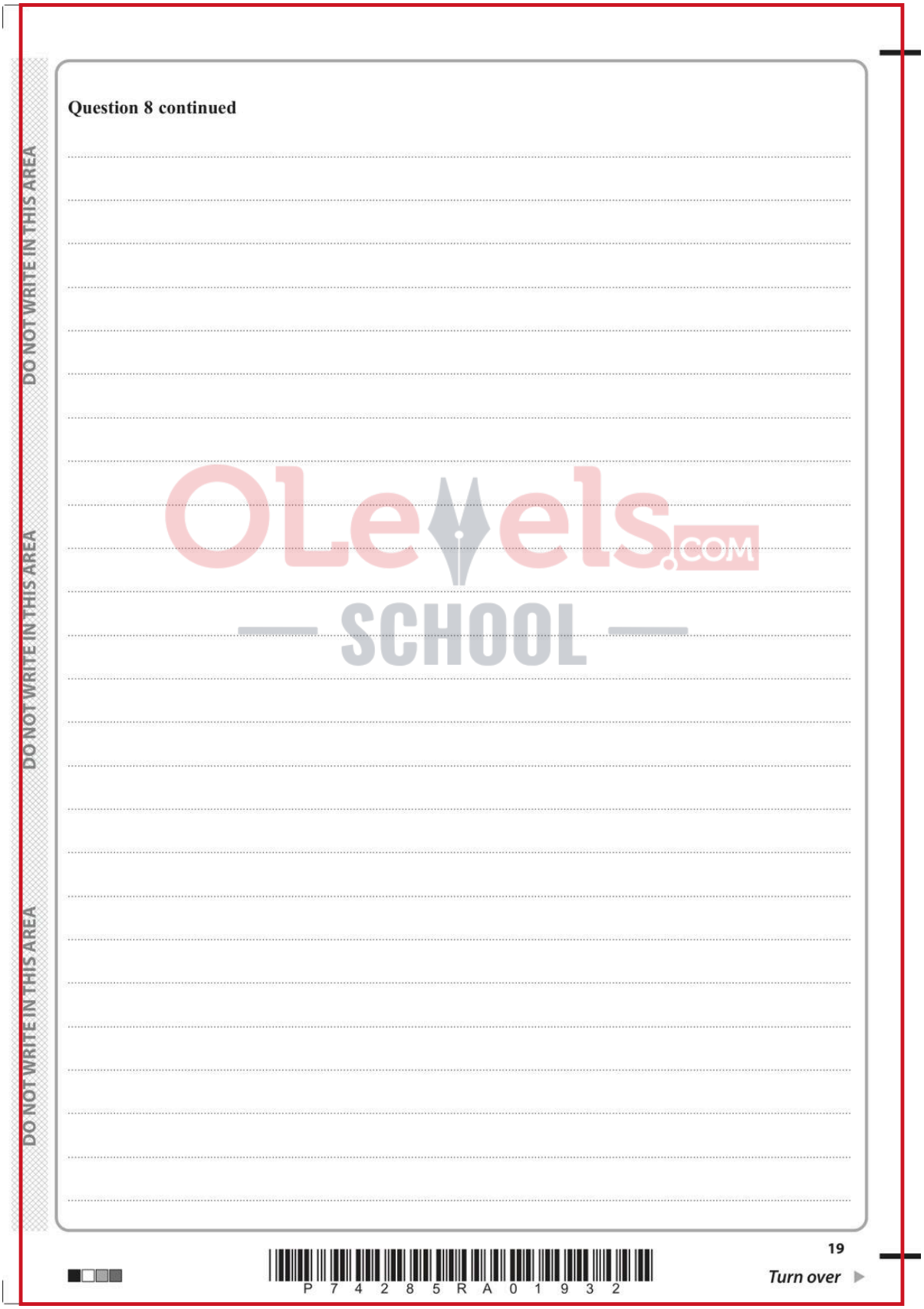
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7. Question 8 · part d · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.



Question 8 continued

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7. Question 8 · part d · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 8 continued

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7. Question 8 · part d · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 8 continued

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(Total for Question 8 is 14 marks)

7. Question 8 · part d · 4 marks · Working and answer

Working and answer

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8. Question 6 · part b.ii · 5 marks

2025 NOV · Paper 02

Further practice · Attempt part b.ii only.

6 (i) A geometric series G has first term $\left(3+\sqrt{3}\right)$ and common ratio r

The second term of G is $\sqrt{3}$

(a) Find, showing all your working, the exact value of r

Give your answer in the form $\frac{p+\sqrt{3}}{q}$ where p and q are integers to be found.

(2)

(b) Explain why G is convergent.

(1)

(ii) A different geometric series H has first term 8 and common ratio 0.6

The sum to n terms of H is S_n and the sum to infinity of H is T

Find, using logarithms, the least value of n such that $T - S_n < 0.12$

(5)

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8. Question 6 · part b.ii · 5 marks

2025 NOV · Paper 02

Further practice · Attempt part b.ii only.

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Question 6 continued

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(Total for Question 6 is 8 marks)

8. Question 6 · part b.ii · 5 marks · Working and answer

Working and answer

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9. Question 7 · part d · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

7 The geometric series G has first term a, common ratio r and nth term u_n

Given that $u_4 = e^{x+2}$ and that $u_7 = e^{\frac{2x+1}{2}}$

(a) show that $r = e^{-\frac{1}{2}}$

(3)

(b) Hence find a in terms of e and x.

(3)

Given that the sum to infinity of G can be written as $\frac{e^p}{e^{\frac{1}{2}} - 1}$

(c) find an expression for p in terms of x.

(3)

Given that $u_{18} > 1.6$ and that x is an integer,

(d) find the least value of x.

(4)

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9. Question 7 · part d · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

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Question 7 continued

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9. Question 7 · part d · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 7 continued

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9. Question 7 · part d · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.



Question 7 continued

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(Total for Question 7 is 13 marks)

9. Question 7 · part d · 4 marks · Working and answer

Working and answer

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10. Question 7 · part b · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

7 A geometric series G has first term a and common ratio r

The sum of the first three terms of G is $\frac{61}{6}$

The sum to infinity of G is $\frac{125}{6}$

(a) (i) Show that $r = \frac{4}{5}$

(ii) Find the value of a

(6)

The sum of the first n terms of G is S_n

Given that $S_n > 19.8$

(b) show that $n \lg\left(\frac{4}{5}\right) < \lg\left(\frac{31}{625}\right)$

(2)

(c) Hence find the least value of n

(2)

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10. Question 7 · part b · 2 marks · Working and answer

Working and answer

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11. Question 7 · part f · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part f only.

7 (a) Use the factor theorem to show that $(4x-1)$ is a factor of

$$f(x) = 64x^3 - 64x^2 + 3$$

(2)

(b) Hence, or otherwise, find the exact roots of the equation

$$f(x) = 0$$

(4)

A geometric series G has first term a and common ratio r
The third term of G is 9 and the sum to infinity of G is 192

(c) Show that $64r^3 - 64r^2 + 3 = 0$

(3)

Given that r is a rational number

(d) write down the value of r

(1)

(e) show that $a = 144$

(2)

The sum to n terms of G is S_n

(f) Using logarithms, find the least value of n such that $S_n > 191.9$

(4)

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11. Question 7 · part f · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part f only.

Question 7 continued

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11. Question 7 · part f · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part f only.

Question 7 continued

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11. Question 7 · part f · 4 marks

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Further practice · Attempt part f only.

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Question 7 continued

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(Total for Question 7 is 16 marks)

11. Question 7 · part f · 4 marks · Working and answer

Working and answer

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12. Question 7 · part c · 5 marks

2023 NOV · Paper 01

Further practice · Attempt part c only.

7 A geometric series G with common ratio r, has first term 16 and third term $\frac{2704}{625}$

(a) Find the two possible values of r

(2)

Given that $r > 0$

(b) find the sum to infinity of G

(2)

The sum to n terms of G is greater than 33

(c) Find, using logarithms, the least possible value of n
Show your working clearly.

(5)

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12. Question 7 · part c · 5 marks

2023 NOV · Paper 01

Further practice · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 9 marks)

12. Question 7 · part c · 5 marks · Working and answer

Working and answer

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13. Question 7 · part c · 5 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

7 The sum of the first n terms of an arithmetic series is A_n where

$$A_n = \sum_{r=1}^{n} (4r + 5)$$

(a) For this arithmetic series, find

(i) the first term,

(ii) the common difference.

(2)

The sum of the first n terms of a geometric series is G_n where

$$G_n = \sum_{r=1}^{n} 4(3)^{r-1}$$

(b) For this geometric series, find

(i) the first term,

(ii) the common ratio.

(2)

(c) Find the value of n for which $A_{14} - 6 = G_n$

(5)

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13. Question 7 · part c · 5 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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13. Question 7 · part c · 5 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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13. Question 7 · part c · 5 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 9 marks)

13. Question 7 · part c · 5 marks · Working and answer

Working and answer

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14. Question 7 · part c · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

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The sum of the first three terms of G is $\frac{61}{6}$

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(ii) Find the value of a

(6)

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(b) show that $n \lg\left(\frac{4}{5}\right) < \lg\left(\frac{31}{625}\right)$

(2)

(c) Hence find the least value of n

(2)

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14. Question 7 · part c · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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14. Question 7 · part c · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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14. Question 7 · part c · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 10 marks)

14. Question 7 · part c · 2 marks · Working and answer

Working and answer

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